

Academic Positions

Columbia University

Founder's Postdoctoral Fellow, Department of Statistics

New York, NY

2025–present

Education

University of Pennsylvania

Ph.D. in Statistics, The Wharton School

Advisor: T. Tony Cai

Philadelphia, PA

2020–2025

Indian Statistical Institute

M.Stat., specialization in Probability

Ranked first; First Division with Distinction

Kolkata, India

2018–2020

Indian Statistical Institute

B.Stat. (Honours)

Ranked first; First Division with Distinction

Kolkata, India

2015–2018

Publications

Author notation: ($\alpha\beta$) Alphabetical ordering. * Equal contribution.

1. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for functional mean estimation under heterogeneous privacy constraints. In press, *Journal of the Royal Statistical Society: Series B*, 2026. [paper]
2. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for nonparametric regression with heterogeneous distributed differential privacy constraints. In press, *Journal of the American Statistical Association*, 2026. [paper]
3. **Abhinav Chakraborty**, Sayak Chatterjee, and Sagnik Nandy. PriME: Privacy-aware membership profile estimation in networks. In press, *SIAM Journal on Mathematics of Data Science*, 2026. [paper]
4. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Yichen Wang. Optimal differentially private ranking from pairwise comparisons. *Journal of the American Statistical Association*, 2026. [paper]
5. ($\alpha\beta$) Arnab Auddy, T. Tony Cai, and **Abhinav Chakraborty**. Minimax and adaptive transfer learning for nonparametric classification under distributed differential privacy constraints. *Journal of the Royal Statistical Society: Series B*, 88(3):903–929, 2026. [paper]
6. **Abhinav Chakraborty**, Arnab Auddy, and T. Tony Cai. When data can't meet: Estimating correlation across privacy barriers. *Advances in Neural Information Processing Systems*, 38, 2025. **Spotlight**. [paper]
7. Ziang Niu*, **Abhinav Chakraborty***, Oliver Dukes, and Eugene Katsevich. Reconciling model-X and doubly robust approaches to conditional independence testing. *The Annals of Statistics*, 52(3):895–921, 2024. [paper]
8. **Abhinav Chakraborty***, Anirban Chatterjee*, and Abhinandan Dalal*. PrIsing: Privacy-preserving peer effect estimation via Ising model. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, PMLR 238:2692–2700, 2024. [paper]

Preprints

1. Lucas Kania*, **Abhinav Chakraborty***, Edward Kennedy, Larry Wasserman, and Sivaraman Balakrishnan. Optimal inference with black-box predictions. arXiv preprint arXiv:2608.10155, 2026. [paper]
2. **Abhinav Chakraborty** and Sagnik Nandy. Transfer learning in high-dimensional clustering: Minimax thresholds and applications in single-cell data. arXiv preprint arXiv:2607.25031, 2026. [paper]
3. **Abhinav Chakraborty** and Subha Maity. The statistical cost of adaptation in multi-source transfer learning. arXiv preprint arXiv:2605.09471, 2026. [paper]

4. **Abhinav Chakraborty***, Yuetian Luo*, and Rina Foygel Barber. Stability and accuracy trade-offs in statistical estimation. arXiv preprint arXiv:2601.11701, 2026. Under major revision at *IEEE Transactions on Information Theory*. [paper]
5. **Abhinav Chakraborty***, Junu Lee*, and Eugene Katsevich. Comparing three learn-then-test paradigms in a multivariate normal means problem. arXiv preprint arXiv:2601.07764, 2026. [paper]
6. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. The cost of adaptation under differential privacy: Optimal adaptive federated density estimation. arXiv preprint arXiv:2512.14337, 2025. [paper]
7. Sayak Chatterjee, Anirban Chatterjee, **Abhinav Chakraborty**, and Bhaswar B. Bhattacharya. Asymptotic normality of subgraph counts in sparse inhomogeneous random graphs. arXiv preprint arXiv:2512.12937, 2025. [paper]
8. **Abhinav Chakraborty***, Jeffrey Zhang*, and Eugene Katsevich. Doubly robust and computationally efficient high-dimensional variable selection. arXiv preprint arXiv:2409.09512, 2024. [paper]
9. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Federated nonparametric hypothesis testing with differential privacy constraints: Optimal rates and adaptive tests. arXiv preprint arXiv:2406.06749, 2024. R&R at *JMLR*. [paper]
10. **Abhinav Chakraborty**, Soumendu Sundar Mukherjee, and Arijit Chakrabarti. High dimensional PCA: A new model selection criterion. arXiv preprint arXiv:2011.04470, 2020. [paper]

Awards and Honors

IMS Travel Award	2025
For the Frontiers in Statistical Machine Learning Workshop.	
Travel Award, Optimization and Statistical Learning Workshop	2025
Columbia University.	
Lawrence D. Brown Best Student Paper Award	2024
The Wharton School, University of Pennsylvania.	
George James Term Award	2022, 2024
Institute of Mathematical Statistics Annual Meeting.	
ISI Alumni Association J. K. Ghosh Memorial Gold Medal	2020
Top scorer in the M.Stat. program.	
P. C. Mahalanobis Gold Medal	2020
Outstanding M.Stat. student; best master's dissertation.	
ISI Alumni Association Mrs. M. R. Iyer Memorial Gold Medal	2018
Top scorer in the B.Stat. program.	
Nikhilesh Bhattacharya Gold Medal	2018
Highest marks in Statistics in the B.Stat. (Hons.) program.	
INSPIRE Scholarship	2015
Awarded to students in the top 1% of the Indian School Certificate Examination.	
Indian National Mathematics Olympiad qualifier	2014
Top-30 candidate in the Regional Mathematics Olympiad, 2013.	

Talks

MCP Conference, Philadelphia, PA	2025
Invited talk: Learn-then-test procedures.	
IISA Conference, Lincoln, NE	2025
Invited talk: Federated transfer learning.	
IMS Annual Meeting, Bochum, Germany	2024
Federated nonparametric regression.	
Ph.D. Student Seminar, University of Pennsylvania	2023
Optimal distributed private estimators.	
IMS Annual Meeting, London, UK	2022
High dimensional PCA.	
P. C. Mahalanobis Gold Medal Presentation, Indian Statistical Institute	2020

Model selection in high dimensional PCA.

D. Basu Memorial Presentation, Indian Statistical Institute

2018

MCMC methods in Bayesian inference.

Teaching and Mentoring

Teaching Assistant, The Wharton School, University of Pennsylvania

2021–2024

Introduction to Probability (STAT 4300); Probability Theory (MATH 6480); Mathematical Statistics (STAT 971); Applied Econometrics (STAT 520).

Ph.D. Peers Program, The Wharton School, University of Pennsylvania

2023

Mentored an incoming Wharton Ph.D. student in navigating the program and transitioning to Penn.

Professional Service

Journal reviewer: *The Annals of Statistics*; *Journal of the American Statistical Association (JASA)*; *IEEE Transactions on Information Theory*; *Journal of the Royal Statistical Society: Series B (JRSSB)*; and *Biometrika*.